Q1. JEE Main 2026 (21 January Shift 2)

Given below are some of the statements about Mn and Mn_2O_7. Identify the correct statements.
A. Mn forms the oxide Mn_2O_7, in which Mn is in its highest oxidation state.
B. Oxygen stabilizes the Mn in higher oxidation states by forming multiple bonds with Mn.
C. Mn_2O_7 is an ionic oxide.
D. The structure of Mn_2O_7 consists of one bridged oxygen.
Choose the correct answer from the options given below :

(1) A, B and D Only
(2) A, B and C Only
(3) A, B, C and D
(4) A, C and D Only

Q2. JEE Main 2026 (21 January Shift 2)

On heating a mixture of common salt and $K_2Cr_2O_7$ in equal amount along with concentrated H_2SO_4 in a test tube, a gas is evolved. Formula of the gas evolved and oxidation state of the central metal atom in the gas respectively are:

(1) $Cr_2O_2Cl_2$ and $+3$
(2) CrO_2Cl_2 and $+5$
(3) CrO_2Cl_2 and $+6$
(4) $Cr_2O_2Cl_2$ and $+6$

Q3. JEE Main 2026 (28 January Shift 2)

Consider the following statements about manganate and permanganate ions. Identify the correct statements.
A. The geometry of both manganate and permanganate ions is tetrahedral.
B. The oxidation states of Mn in manganate and permanganate are +7 and +6, respectively.
C. Oxidation of Mn (II) salt by peroxodisulphate gives manganate ion as the final product.
D. Manganate ion is paramagnetic and permanganate ions is diamagnetic.
E. Acidified permanganate ion reduces oxalate, nitrite and iodide ions.
Choose the correct answer from the options given below :

(1) A, C and D Only
(2) A and D Only
(3) A, D and E Only
(4) A, B and C Only

Q4. JEE Main 2026 (24 January Shift 2)

" X " is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in +6 oxidation state in " X ". The color of the potassium salt of X is

(1) purple
(2) orange
(3) yellow
(4) green

Q5. JEE Main 2026 (22 January Shift 2)

Given below are two statements :

Statement I: The first ionization enthalpy of Cr is lower than that of Mn .

Statement II: The second and third ionization enthalpies of Cr are higher than those of Mn.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is false but Statement II is true

(4) Statement I is true but Statement II is false

Q6. JEE Main 2026 (22 January Shift 2)

Among the following oxides of 3 d elements, the number of mixed oxides are ____。

$Ti_2O_3, V_2O_4, Cr_2O_3, Mn_3O_4, Fe_3O_4, Fe_2O_3, Co_3O_4$

Q7. JEE Main 2026 (22 January Shift 1)

A first row transition metal (M) does not liberate H_2 gas from dilute HCl.1 mol of aqueous solution of MSO_4 is treated with excess of aqueous KCN and then H_2 S(g) is passed through the solution. The amount of MS (metal sulphide) formed from the above reaction is ____ mol.

(1) 0 (2) 3 (3) 1 (4) 2

Q8. JEE Main 2026 (23 January Shift 2)

The oxidation state of chromium in the final product formed in the reaction between KI and acidified $K_2Cr_2O_7$ solution is:

(1) +6 (2) +3 (3) +4 (4) +2

Q9. JEE Main 2026 (21 January Shift 1)

$MnO_4{}^{2-}$, in acidic medium, disproportionates to :

(1) $MnO_4{}^-$and MnO_2

(2) Mn_2O_7 and MnO_2

(3) Mn_2O_7 and MnO

(4) $MnO_4{}^-$ and MnO

Q10. JEE Main 2026 (21 January Shift 1)

Consider the following reactions:

$NaCl + K_2Cr_2O_7 + H_2SO_4 \rightarrow A + KHSO_4 + NaHSO_4 + H_2O$

$A + NaOH \rightarrow B + NaCl + H_2O$

$B + H_2SO_4 + H_2O_2 \rightarrow C + Na_2SO_4 + H_2O$

In the product ' C ', ' X ' is the number of O_2^{2-} units, ' Y ' is the total number oxygen atoms present and ' Z ' is the oxidation state of Cr. The value of $X + Y + Z$ is ____.

Q11. JEE Main 2026 (28 January Shift 1)

Given below are two statements:

Statement I: The number of pairs, from the following, in which both the ions are coloured in aqueous solution is 3.
$[Sc^{3+}, Ti^{3+}]$, $[Mn^{2+}, Cr^{2+}]$, $[Cu^{2+}, Zn^{2+}]$ and $[Ni^{2+}, Ti^{4+}]$

Statement II: Th^{4+} is the strongest reducing agent among Th^{4+}, Ce^{4+}, Gd^{3+} and Eu^{2+}.

In the light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are false
(2) Both Statement I and Statement II are true
(3) Statement I is false but Statement II is true
(4) Statement I is true but Statement II is false

ANSWERS AND SOLUTIONS

1. (1)	**2.** (3)	**3.** (2)	**4.** (4)	**5.** (4)	**6.** 3	**7.** (1)	**8.** (2)
9. (1)	**10.** 13	**11.** (1)					

1. **(1)** Analyzing each statement:

Statement A: In Mn_2O_7, using oxidation state calculation: 2(Mn) + 7(O^{-2}) = 0, so Mn oxidation state = +7. This is indeed the highest oxidation state for Mn. TRUE.

Statement B: Oxygen stabilizes Mn in +7 oxidation state through multiple bonding (Mn=O double bonds and bridging), reducing electron density effectively. TRUE.

Statement C: Mn_2O_7 is primarily a covalent oxide with significant Mn-O covalent bonding character. The oxide is not ionic in nature. FALSE.

Statement D: The structure of Mn_2O_7 contains bridged oxygen atoms (Mn-O-Mn linkages) along with terminal Mn=O bonds. TRUE.

Statements A, B, and D are correct while C is incorrect. The answer is option (1): A, B and D Only.

2. **(3)** When a mixture of NaCl and K2Cr2O7 is heated with concentrated H2SO4, the dichromate ion oxidizes chloride ions. The reaction produces chromyl chloride (CrO2Cl), which is a volatile deep-red colored gas. The dichromate ion contains chromium in the +6 oxidation state. In the reaction, the dichromate is reduced and chloride is oxidized, but the chromium in the product remains in the +6 oxidation state as CrO2Cl (or as $CrO2^+$ in ionic form). The gas evolved is CrO2Cl with chromium in the +6 oxidation state

3. **(2)** Manganate ion → MnO_4^{2-}
Permanganate ion → MnO_4^-
(A) Both are tetrahedral (d^3s hybridisation) TRUE
(B) MnO_4^- has Mn in +7 oxidation state; MnO_4^{2-} has Mn in +6 oxidation state. The statement says +7 for manganate and +6 for permanganate, which is reversed. FALSE
(C) Oxidation of Mn^{2+} by peroxodisulphate ($S_2O_8^{2-}$) gives permanganate ion (MnO_4^-), not manganate ion. FALSE
(D) MnO_4^- is diamagnetic; MnO_4^{2-} is paramagnetic. TRUE
(E) Acidified permanganate ion oxidises (not reduces) oxalate, nitrite and iodide ions. FALSE
Correct statements: A and D only.

4. **(4)** The lightest element of group 7 (considering transition metals) is manganese (Mn).
An oxoanion with Mn in +6 oxidation state is manganate, MnO_4^{2-}.
In manganate, Mn exhibits +6 oxidation state (where oxygen is -2, so Mn + 4(-2) = -2, giving Mn = +6).
The potassium salt K_2MnO_4 (potassium manganate) is green in color.
This is distinct from permanganate $KMnO_4$ (where Mn is +7) which is purple.
The characteristic green color of manganate solutions and salts makes this the answer.

5. **(4)** Statement I:
The electronic configuration of Cr is $[Ar]3d^5 4s^1$ and Mn is $[Ar]3d^5 4s^2$.
For Cr, the first electron is removed from $4s^1$, while for Mn, it is removed from a stable fully filled $4s^2$ subshell.
Additionally, Mn has a higher nuclear charge.
Thus, the first ionization enthalpy (IE_1) of Cr (653 kJ/mol) is lower than that of Mn (717 kJ/mol).
Statement I is true.

Statement II:
For the second ionization enthalpy (IE_2), Cr^+ has configuration $[Ar]3d^5$ (stable half-filled) and Mn^+ has $[Ar]3d^5 4s^1$.
Removing an electron from the stable $3d^5$ configuration of Cr^+ requires more energy than removing it from the $4s^1$ of Mn^+.
Thus, IE_2 of Cr (1592 kJ/mol) is higher than Mn (1509 kJ/mol).
For the third ionization enthalpy (IE_3), Cr^{2+} is $[Ar]3d^4$ and Mn^{2+} is $[Ar]3d^5$.
Removing an electron from the stable half-filled $3d^5$ configuration of Mn^{2+} requires significantly more energy than from the $3d^4$ of Cr^{2+}.
Thus, IE_3 of Mn (3248 kJ/mol) is higher than Cr (2987 kJ/mol).
Since IE_3 of Cr is lower than Mn, Statement II is false.

6. **(3)** Mixed oxides contain the same element in two different oxidation states.
Analyzing each oxide:
Ti_2O_3 contains Ti(+3) only - simple oxide.
V_2O_4 contains V(+4) only - simple oxide.
Cr_2O_3 contains Cr(+3) only - simple oxide.
Mn_3O_4 can be written as $MnO \cdot Mn_2O_3$, containing both Mn(+2) and Mn(+3) - mixed oxide.
Fe_3O_4 can be written as $FeO \cdot Fe_2O_3$, containing both Fe(+2) and Fe(+3) - mixed oxide.
Fe_2O_3 contains Fe(+3) only - simple oxide.
Co_3O_4 can be written as $CoO \cdot Co_2O_3$, containing both Co(+2) and Co(+3) - mixed oxide.
Therefore, three mixed oxides are present: Mn_3O_4, Fe_3O_4, and Co_3O_4.

7. **(1)** The first row transition metal (M) that does not liberate H_2 gas from dilute HCl is Copper (Cu), as it has a positive reduction potential ($E^\circ_{Cu^{2+}/Cu} = +0.34$ V).

When 1 mol of $CuSO_4$ reacts with excess aqueous KCN, it first forms $Cu(CN)_2$, which is unstable and decomposes to $CuCN$ and cyanogen gas $(CN)_2$.
$2CuSO_4 + 4KCN \rightarrow 2Cu(CN)_2 + 2K_2SO_4$
$2Cu(CN)_2 \rightarrow 2CuCN + (CN)_2$
The $CuCN$ then dissolves in excess KCN to form a highly stable soluble complex, potassium tetracyanocuprate(I):
$CuCN + 3KCN \rightarrow K_3[Cu(CN)_4]$
The complex $[Cu(CN)_4]^{3-}$ is so stable that its dissociation constant is extremely low. When H_2S gas is passed through this solution, the concentration of Cu^+ ions is insufficient to exceed the solubility product of Cu_2S. Therefore, no precipitate of metal sulphide is formed.
The amount of MS (or M_2S) formed is 0 mol.

8. **(2)** In the reaction between KI and acidified $K_2Cr_2O_7$, Cr in the dichromate ion exists in +6 oxidation state. Iodide ions are oxidized to I_2 while chromium is reduced. The reduction product is Cr^{3+} ions, which form chromium(III) salts like $Cr_2(SO_4)_3$ or coordination complexes. Therefore, the oxidation state of chromium in the final product is +3.

9. **(1)** MnO_4^{2-} (manganate) has Mn in +6 oxidation state.
In acidic medium, it undergoes disproportionation where Mn is simultaneously oxidized and reduced:
Oxidation: $+6 \rightarrow +7$ (forms MnO_4^-, permanganate)
Reduction: $+6 \rightarrow +4$ (forms MnO_2)
Reaction: $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

10. **(13)** Reaction 1: $NaCl + K_2Cr_2O_7 + H_2SO_4 \rightarrow CrO_2Cl_2$ (A, chromyl chloride)
Reaction 2: $CrO_2Cl_2 + 4NaOH \rightarrow Na_2CrO_4 + 2NaCl + 2H_2O$, B = Na_2CrO_4
Reaction 3: $Na_2CrO_4 + H_2SO_4 + H_2O_2 \rightarrow CrO_5 + Na_2SO_4 + H_2O$
C = CrO_5 or $CrO(O_2)_2$ (chromium peroxide, blue)
Structure: one oxo group and two peroxide (O_2^{2-}) units
X = number of O_2^{2-} units = 2
Y = total oxygen atoms = 5
Z = oxidation state of Cr: $O^{2-}(-2) + 2 \times O_2^{2-}(-4) = -6$, so Cr = +6
$X + Y + Z = 2 + 5 + 6 = 13$

11. **(1)** [Sc^{3+}, Ti^{3+}] has Sc^{3+} colorless (no d-electrons) and Ti^{3+} colored;
[Mn^{2+}, Cr^{2+}] has Mn^{2+} colorless (d^5 half-filled, no d-d transitions) and Cr^{2+} colored;
[Cu^{2+}, Zn^{2+}] has Cu^{2+} colored and Zn^{2+} colorless (d^{10} filled);
[Ni^{2+}, Ti^{4+}] has Ni^{2+} colored and Ti^{4+} colorless (no d-electrons).
Zero pairs have both ions colored. Statement I is FALSE.

Th^{4+} is not a reducing agent. Statement II is FALSE.

Both statements are false.